QWeatherRiskF

Verification and fix plan: cotton 2019, oilseeds, minor pulses

Purpose. Decide, with evidence, what is wrong with these three series, how to fix each, and what can be claimed about the data source. No dataset has been changed. All external figures were taken from published sources listed at the end and should be opened and confirmed before they are quoted.

1. Method: how to verify a series when there is no second dataset

When a series looks wrong, test it against independent official statistics for the same state and year (triangulation), using three tests: (a) **ratio test**, our value divided by the official value; (b) **stability test**, whether the series jumps between years while area is steady; (c) **internal consistency**, such as yield = production / area and aggregate = sum of members. The decision rule is: a ratio near 1 within the normal disagreement between official sources passes; a ratio that is constant across years and districts and equals a round unit factor (like sugarcane x10) is a unit error and is corrected by that factor; a ratio that is not constant is a data error that cannot be fixed by scaling, so the value is replaced from the official source or set to missing.

2. Cotton 2019: confirmed error in production and yield (area is correct)

Year	Dataset area (000 ha)	Official area (000 ha)	Dataset production (000 t)	Official production (000 t)	Ratio (dataset / official)	Dataset yield / official yield (kg/ha)
2015	667	666	327	404	0.81	490 / 606
2016	473	472	282	323	0.87	596 / 684
2017	647	644	374	348	1.07	578 / 541
2018	619	620	268	253	1.06	433 / 409
2019	657	657	80	426	0.19	122 / 649

Official area and production are Andhra Pradesh state series from the Directorate of Economics and Statistics, Ministry of Agriculture, as tabulated by ICAR-CICR [3]. Production in lakh bales of 170 kg was converted to thousand tonnes (x 17).

- **Area matches the official series in every year** (ratio 1.00), and in 2019 for each of the five main districts: Kurnool 275, Guntur 182, Anantapur 52, Krishna 48, Prakasam 46 thousand ha, identical to the AP district table [4]. So the area, district and year labels are correct.
- **Production for 2015-2018 is within -19% to +7% of the official series,** which is inside the disagreement between official sources themselves (AP 2015-16 appears as 18.17 lakh bales in one AP series [4] and 23.75 in another [3][5]). That explains why published reports do not match each other or the dataset exactly.
- **Production for 2019 is only 19% of the official figure** (80 against 426 thousand tonnes; official 25.08 lakh bales, the highest of the five years). In the five main districts the ratio is 0.18-0.22 (see table below), so the error is a large understatement, not a real fall.
- **It is not a clean unit error.** The ratio varies from 0.18 to 0.24 across districts (mean 0.19, SD 0.016 over the five main districts), and production is stored as whole thousand tonnes, so small districts show rounding. Multiplying by about 5.2 would not be defensible.

District (2019)	Dataset production (000 t)	Official production (000 t)	Ratio
Kurnool	28	153.7	0.18
Guntur	30	167.6	0.18
Anantapuramu	3	15.5	0.19
Krishna	7	37.7	0.18
Prakasam	5	22.8	0.22
Other 8	7	29.1	0.24

Official district values: AP Agricultural Statistics at a Glance as reported by ANGRAU [4]; "Other 8" is the combined remainder (1.71 lakh bales).

Recommended fix (for decision). (1) Obtain the official district-wise 2019-20 cotton production from AP Directorate of Economics and Statistics or the Agricultural Statistics at a Glance district tables and replace the 2019 production, then recompute yield = production / area x 1000. For the five main districts the values in the table above can be used (x 17), but the eight small districts are only available as one total, so the district table is needed. (2) If official district

values cannot be obtained, set 2019 cotton production and yield to missing, keep the (correct) area, and exclude 2019 from cotton modelling. Do not rescale by a constant.

3. Oilseeds aggregate: composition is wrong, so use the individual crops

- In 35 of 65 district-years the oilseeds total is far from the sum of its eight member crops (7 districts, every year), while it matches exactly in the other 6.
- **Evidence for the cause.** The excess area is largest exactly where official data place oil palm: West Godavari (excess 52-74 thousand ha, official oil palm 78-94 thousand ha) and Krishna (11-23 against 15-19) [6]. Across the 8 oil-palm districts and 5 years, excess area and official oil palm area correlate (Spearman 0.74, n = 40). The implied yield of the excess production is 15,000-25,000 kg/ha in West Godavari and Krishna, which is the range of oil palm fruit-bunch yield (about 12-26 tonnes per hectare in a survey of bearing plantations [6]) and far above any seed oilseed yield.
- **The match is not clean:** the excess in East Godavari, Srikakulam and Visakhapatnam is 2-4 times oil palm area in some years and near zero in others, and Nellore and Vizianagaram go negative in 2018-19. Probably other perennial oil crops (e.g. coconut) are included in some years, but this is unconfirmed. The inconsistent composition by year is itself enough to reject the aggregate.
- **Decision.** Agree with your proposal: **drop the oilseeds aggregate from modelling** and use the individual crops. Reasons: it double counts its members, it includes crops outside the list, and its composition changes between years. Keep it in the long file, flagged, for traceability.

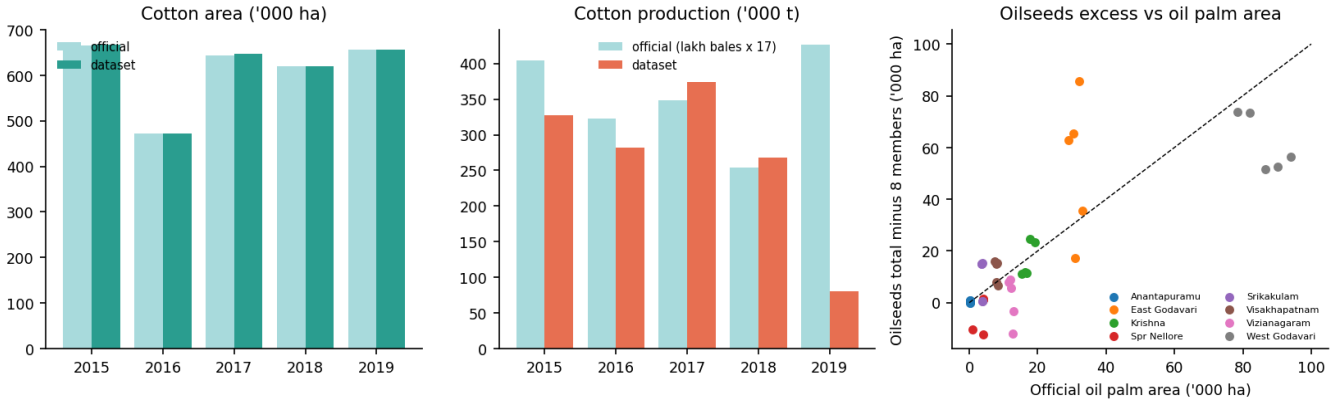


Figure 1. Cotton area and production against official series (left, centre); oilseeds excess area against official oil palm area by district-year (right; dashed line = 1:1).

4. Minor pulses 2018-19: zeros are placeholders, not real absence

- The dataset shows 600-740 thousand ha of minor pulses in 2015-17, then exactly 0 area, production and yield in 12 of 13 districts in 2018 and 2019 (only Anantapuramu has small values).
- **These crops did not disappear.** Green gram, black gram and horse gram are rabi pulses that the AP Commissionerate of Agriculture reports growing, with crop stages and pest incidence, in the rabi 2018-19 and 2019-20 season and crop coverage reports [7]. The ANGRAU outlook gives about 4 lakh ha of black gram and 1 lakh ha of green gram in the state in 2021-22 [8], the same order as our 2015-17 values.
- **Minor pulses is a residual category** (pulses other than chickpea and pigeonpea). Its members are not in the dataset, so it cannot be checked by summation. A category that switches to zero for two years is a reporting or definition change, not a unit error, so no scaling can fix it.
- **Decision.** Agree with your proposal: **exclude minor pulses from modelling.** If it is kept in the long file, treat 2018-19 zeros as missing, not zero. If these pulses matter, rebuild them as individual crops (green gram, black gram, horse gram) from the AP district tables.

5. What can be claimed about ICRISAT as the source

- ICRISAT's District Level Database is a **compiled secondary dataset**. ICRISAT states that it was built from published government sources and archival records, and the portal covers 1966-2017 in the descriptions I found [9][10]. The primary official sources are the Directorate of Economics and Statistics (Ministry of Agriculture) and the state agriculture departments (for AP, DES and the Commissionerate of Agriculture).
- The data are open (CC BY 4.0) but supplied "as is" [11]. The ICRISAT documentation has a section on data clarifications and anomalies and asks users to report inconsistencies [12]. So the accurate wording is: *"crop area, production and yield from the ICRISAT District Level Database, compiled from government statistics, validated against Andhra Pradesh state and national official series"*. It is better not to call ICRISAT itself the official statistical authority.
- **Check where the 2018-19 rows came from.** The published coverage I found ends in 2017, while the breaks in this dataset (sugarcane units, minor pulses, oilseeds, cotton 2019) all start in 2018. That pattern fits 2018-19 being appended from a different source or processing step. If a newer DLD release or another source supplied those two years, its release notes and definitions are the first thing to check.

6. Recommended next steps (before any dataset update)

- Confirm the source and processing of the 2018-19 rows with whoever prepared them.
- Run the same tie-out against official series for every crop in 2018-19 (not only these three) and record the ratios; two checks already show that 2018-19 values cannot be assumed correct.
- Obtain the AP district-wise tables (Agricultural Statistics at a Glance, 2018-19 and 2019-20) to replace cotton 2019 production and to validate the other crops.
- Then apply the decisions: drop aggregates from the modelling set, correct or blank cotton 2019, flag minor pulses 2018-19 as missing.

References (open each link and confirm before quoting)

#	Source	Link
[1]	ICRISAT District Level Data (DLD) portal: area, production and yield file group; clarifications and anomalies	https://data.icrisat.org/dld/src/gdp.html
[3]	ICAR-CICR: state-wise cotton area, production and productivity 2014-15 to 2023-24 (source: DES, Ministry of Agriculture)	https://icar-cicr.org.in/datasets-tables/
[4]	ANGRAU Cotton Outlook Report, Jan-Dec 2021: AP area and production and inter-district comparison 2019-20 (source: AP Agricultural Statistics at a Glance)	https://angrau.ac.in/downloads/AMIC/OutlookReports/2021/4-COTTON_January%20to%20December,%202021.pdf
[5]	PIB, Ministry of Textiles (6 Dec 2019): cotton production and productivity, Andhra Pradesh 2015-16 to 2018-19	https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1595239®=48⟨=2
[6]	Reddy et al. (2022), Economic analysis of oil palm cultivation in Andhra Pradesh, Asian J. Agric. Ext. Econ. Sociol. 40(9): district-wise oil palm area 2015-16 to 2020-21 (AP Horticulture Department); yields per hectare	https://ageconsearch.umn.edu/record/367049/files/sciencedomain%2C%2BReddy4092022AJAEES88153.pdf
[7]	Government of AP, Season and Crop Coverage Reports, Rabi 2018-19 and 2019-20 (green gram, black gram, horse gram)	https://apagrisnet.gov.in/2019/weekly/October/weekly_report_(Rabi)_11_01-01-20.pdf
[8]	ANGRAU outlook reports on green gram and black gram (AP area, 2021-22)	https://angrau.ac.in/downloads/AMIC/OutlookReports/2022/7-Blackgram%20Outlook%20Report%20-%20January%20to%20December%202022.pdf
[9]	ICRISAT Dataverse record: District Level Data and Dashboard (1966-2017, compiled from government sources)	https://krishi.icar.gov.in/ohs-2.3.1/index.php/record/view/662239
[10]	FEWS NET India data book: ICRISAT database covers 1966-2017, compiled from official statistics	https://help.fews.net/fde/india-data-book
[11]	Data card for the ICRISAT DLD (license CC BY 4.0; provided as is)	https://github.blesdmm.dns-dynamic.net/WUR-AI/AgML-CY-Bench/blob/main/data_preparation/crop_statistics_IN/README.md
[12]	ICRISAT VDSA DLD documentation: data clarifications and anomalies, data sources	https://vdsa.icrisat.org/Include/document/all-apportioned-web-document.pdf